MONISHKUMAR E M

Software Engineering candidate with hands-on experience in full-stack development, AI/ML, Generative AI, backend systems, and Data Structures & Algorithms. Experienced in building production-oriented applications using Python, C++, React, Node.js, FastAPI, PostgreSQL, LLMs, and RAG pipelines.

Portfolio: https://www.monishdev.online/

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EDUCATION

BTech in Information Technology — **Aug 2023 - Present**
College of Engineering Guindy, Anna University, Chennai. — **CGPA : 8.20**

HSC (2023)
Saradha Higher Matric Secondary School, Gingee, Villupuram district — **96.17 %**

EXPERIENCE

AstraZeneca – Summer Intern — **May 2026 – Jul 2026**
Chennai, India

- Measured 85-90% accuracy of a Python/FastAPI RAG assistant over 100+ enterprise documents (20-30 pages).
- Automated daily and monthly ServiceNow report ingestion into PostgreSQL for SQL comparison across 8 tables.
- Developed an LLM-powered NL2SQL pipeline over 8 enterprise tables, returning query results in 10-30 seconds.
- Collaborated with cross-functional teams to deliver proof-of-concept systems supporting reporting workflows.

PROJECTS

Enterprise AI Assistant for Company Documents *(Internship Project)* — **AstraZeneca**

- Architected an enterprise assistant using RAG across 100+ internal documents (20-30 pages each).
- Enabled semantic search across PDFs, Excel files, and reports through chunking, embeddings, and vector retrieval.
- Deployed Large Language Models to generate responses with 85-90% accuracy from enterprise knowledge bases.
- Replaced manual document search with instant semantic retrieval, cutting information lookup effort.

Tech Stack: *Python, FastAPI, Ollama, RAG, Vector Database, PostgreSQL, LLMs, LangChain*

ServiceNow Data Analytics Platform *(Internship Project)* — **AstraZeneca**

- Delivered an analytics platform spanning 8 PostgreSQL tables sourced from ServiceNow datasets.
- Automated ingestion of daily and monthly ServiceNow reports into PostgreSQL to streamline comparisons.
- Developed an NL2SQL pipeline using LLMs to generate optimized SQL from natural-language questions.
- Delivered responses in 10-30 seconds per query, with automatic chart and graph generation.
- Supported analytical queries across Incident, Change, Ctask, IDM Task, CMDB, and SCTask tables.

Tech Stack: *Python, FastAPI, LLMs (Ollama), PostgreSQL, SQLAlchemy, Pandas, Matplotlib, Plotly*

SpareChange GitHub — *Full Stack Development (PERN + ML Integration)*

- Piloted a full-stack savings app with 10 test users, rounding daily spend into savings, investments, or donations.
- Established REST APIs with authentication and role-based logic, using a ReactJS frontend and Node.js backend.
- Modelled a PostgreSQL database schema for daily transaction tracking.
- Incorporated an ML-based recommendation system achieving 90% allocation accuracy.
- Ranked in the Top 10 projects department-wide across the entire IT department.

Ingre2Recipe GitHub — *ML*

- *Achieved 95% ingredient match accuracy across 250+ ingredients using custom parsing and matching algorithms.*
- *Integrated ML-based recommendation logic for recipe suggestions.*
- *Structured a modular architecture designed to scale beyond 250+ ingredients and future datasets.*

TECHNICAL SKILLS

Programming Languages	: C++, C, Python, Java, SQL, JavaScript
Testing & Quality	: Debugging, Requirements Analysis, LLM/ML Accuracy Evaluation, Technical Documentation
API & Backend	: REST APIs, FastAPI, Node.js, Express.js
Databases & Data Processing	: PostgreSQL, MongoDB, SQLAlchemy, Pandas
CS Fundamentals	: Data Structures & Algorithms, Object-Oriented Programming, DBMS
AI/ML	: Machine Learning, LLMs, RAG, LangChain, Ollama
Tools	: Git, GitHub, VS Code

ACHIEVEMENTS AND CERTIFICATIONS

- Coordinator in NSS - Coordinated Village Camp at Puliyambakkam, Kanchipuram District, *led community outreach, organised cleanliness drives, and contributed to rural development initiatives.*
- Google Cybersecurity Professional Certificate
- Google Cloud Engineering Certificate
- IBM DevOps and Software Engineering Professional Certificate
- NPTEL course - *Python for Data Science(85%)*
- *Completed three LearnTube courses - C Programming, C++ Programming, Data Structures and Algorithms*